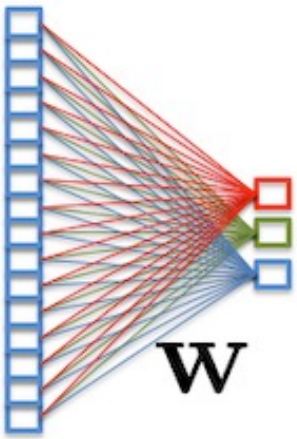
AI and Data Analytics

3.2 Reinforcement Learning



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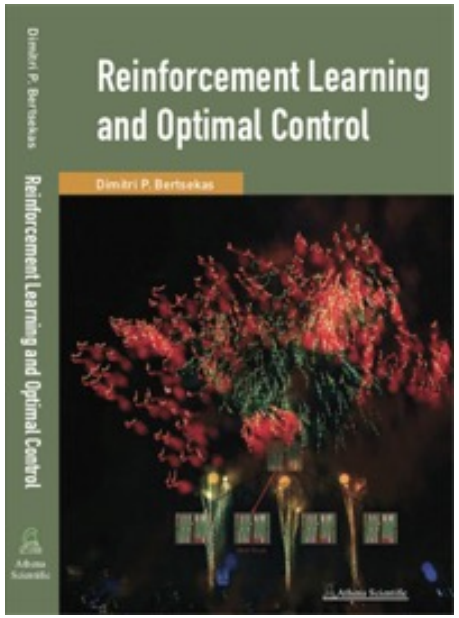
Literature

This presentation is largely inspired by the open-IA Tutorial on RL:

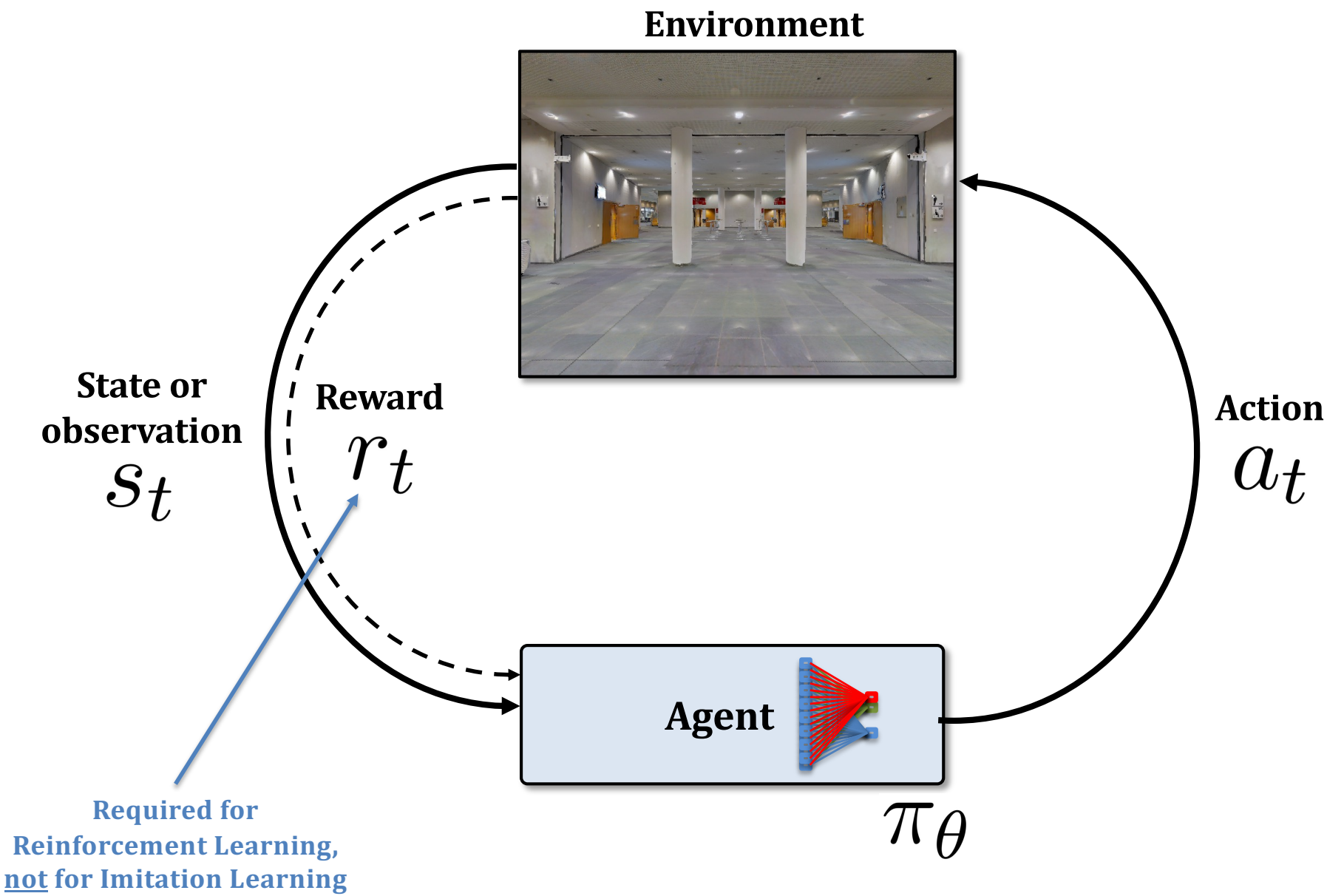
`https://spinningup.openai.com/en/latest/spinningup/rl_intro.html`



A lengthier introduction into Reinforcement learning:

Dimitri P. Bertsekas, "Reinforcement Learning and Optimal Control », 2019.

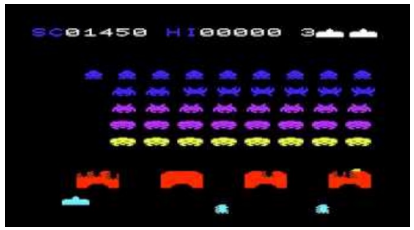
Interacting with an "environment"



States

An environment is in a given state s_t at any instant t.

Fully observable problems — the state is observable to the agent at any time, e.g. board games (chess, Go), 2D computer games with visibe full screen (Pacman).



Partially observable problems — only an observation is returned to the agent at each instant, not the full state. Examples: robot navigation; 3D computer games with ego-centric view.

Actions

At each time t, the agent takes an action a_t from an action space $\mathcal{A}$.

Discrete action spaces

- Chess: $\mathcal{A} = \{$*a1*,*Qf3*,*Ra1*,*bf3*,$\ldots\}$.
- Go: $\mathcal{A} = [1\ldots19] \times\ [1\ldots19]$.
- Dialogue: $\mathcal{A} =$ words or characters.

Continuous action spaces

- Robotic arm control: $\mathcal{A} =$motor commands
- Drone (UAV) control: $\mathcal{A} =$ motor commands

Policies

The policy π of an agent is a function which

- chooses an action given that the agent is in a current state;
- or, in the stochastic case, it assigns a probability to an action:

$$\pi(a_t|s_t)$$

Policies are generally parametrized by a vector θ and implemented as a neural network, denoted as π_θ.

The reward function

After taking action a_t in state s_t, the agent is rewarded with a scalar reward r_t:

$$r_t = R\left(s_t, a_t, s_{t+1}\right)$$

or, simpler:

$$r_t = R\left(s_t, a_t\right)$$

We are most often interested in the accumulation of the reward over time using a horizon T:

$$R(\tau) = \sum_{t=0}^{T} \gamma^t r_t$$

We discount rewards which are further away in the future by a factor γ.

Markov Decision Processes

Mathematically, the situation we just presented is formalized as a Markov Decision Process (MDP)

$$M = \{\mathcal{S}, \mathcal{A}, P, R, p_0\}$$

where

- $\mathcal{S}$ is the set of states.
- $\mathcal{A}$ is the action space.
- $P(s_{t+1}|s_t, a_t)$ is a probability distribution governing state transitions.
- $R(s_t, a_t, s_{t+1})$ is the reward function.
- p_0 is the distribution of the initial state s_0.

MDPs, like Markov Chains, satisfy the **Markov property**: the outcome depends only on the current state, and not on the full history.

Trajectories

A trajectory τ is a sequence of states s_t and actions a_t taken by an agent following a policy. It can be a assigned a probability:

$$P(\tau|\pi) = \rho_0\left(s_0\right) \prod_{t=0}^{T-1} P\left(s_{t+1}|s_t, a_t\right) \pi\left(a_t|s_t\right)$$

Given a trajectory, this probability only depends on the environment (P of the MDP).

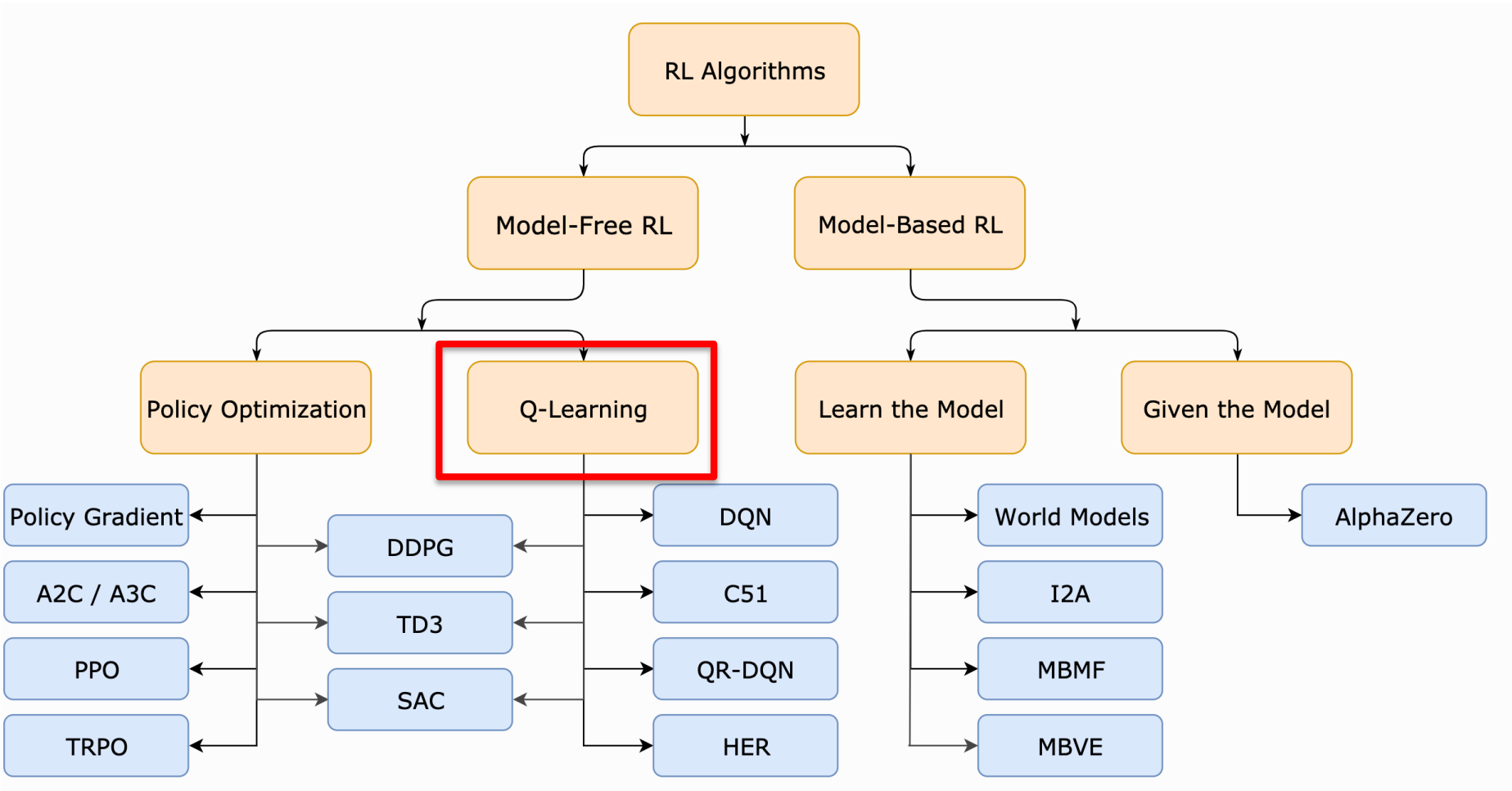
What is the objective?

Our goal is to learn a policy π_θ which optimizes the cumulative reward:

$$\begin{aligned} J(\pi_\theta) &= \mathop{\mathbb{E}}_{\tau \sim \pi_\theta} [R(\tau)] \\ &= \mathop{\mathbb{E}}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} r_{t+1} | \tau \right] \end{aligned}$$

$\Rightarrow$ we want to take actions, which are not only of high reward immediately, but of long term **value**!

Types of reinforcement learning algorithms



https://spinningup.openai.com/en/latest/spinningup/rl_intro2.html

Value functions

The **Value** of state s_t, given a policy π_θ, is given as the expected cumulative return when following the given policy from this state:

$$V^\pi(s) = \mathop{\mathbb{E}}_{\tau \sim \pi} [R(\tau) | s_0 = s]$$

The optimal value function is obtained with the optimal policy:

$$V^*(s) = \max_\pi \mathop{\mathbb{E}}_{\tau \sim \pi} [R(\tau) | s_0 = s]$$

The **Action-value function**:

$$Q^\pi(s, a) = \mathop{\mathbb{E}}_{\tau \sim \pi} [R(\tau) | s_0 = s, a_0 = a]$$

And it's optimal version:

$$Q^*(s, a) = \max_\pi \mathop{\mathbb{E}}_{\tau \sim \pi} [R(\tau) | s_0 = s, a_0 = a]$$

Q-Learning

Q-Learning does not directly learn a policy π, but attemps in estimating the optimal Q-function $Q^*(s, a)$.

The optimal policy is then the one which selects the optimal action a^* in state s as follows:

$$a^*(s) = \arg\max_a Q^*(s, a)$$

$\Rightarrow$ we need to estimate $Q*(s, a)$.
We only have access to the immediate reward r_t at each instant.

Q-Learning

For a given policy π, we can estimate the Q-function with the **Bellman equation**:

$$Q^{\pi}(s,a) = \mathop{\mathbb{E}}_{s' \sim P}\left[r(s,a) + \gamma \mathop{\mathbb{E}}_{a' \sim \pi}\left[Q^{\pi}\left(s',a'\right)\right]\right]$$

But we are interested in the Bellmann equation for the **optimal Q-function**:

$$Q^{*}(s,a) = \mathop{\mathbb{E}}_{s' \sim P}\left[r(s,a) + \gamma \max_{a'} Q^{*}\left(s',a'\right)\right]$$

Q-Learning: value iteration

We update the current estimate of $Q^*(s,a)$ at each time step: The agent is in state s_t, performs an action a, gets a new state s' and a reward $r(s,a)$.

Discrete state and action spaces: we can keep the Q-function in a table and update it:

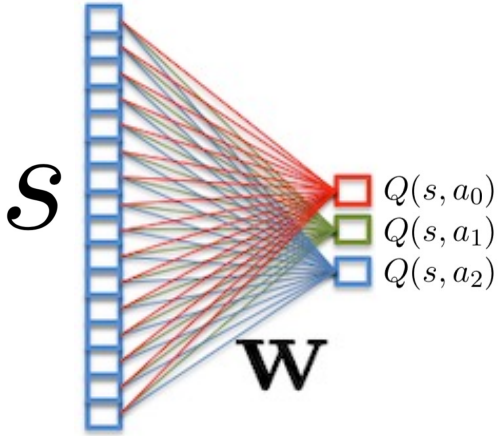
$$Q(s,a)^{[t+1]} = (1-\nu)Q^*(s,a)^{[t]} + \nu \left[r(s,a) + \gamma \max_{a'} Q\left(s', a'\right) \right]$$

where ν is a learning rate.

Q-Learning: value iteration

Continuous state spaces: we approximate the Q-function with a neural network:

$$f(s,\theta) = \boldsymbol{q} = \begin{bmatrix} Q(s,a_0) \\ Q(s,a_1) \\ Q(s,a_2) \\ \vdots \\ Q(s,a_N) \end{bmatrix}$$



We minimize a loss between the old value and the new one:

$$\mathcal{L}\left(Q(s,a),\ r(s,a) + \gamma \max_{a'} Q(s',a')\right)$$

and learn by gradient descent.

Exploration vs. exploitation

In order to learn a reliable and useful value function, the agent needs to explore the state space and can't always take the action which is **currently** estimated as optimal $\Rightarrow$ Exploration/exploitation trade-off.

ϵ-greedy strategy:

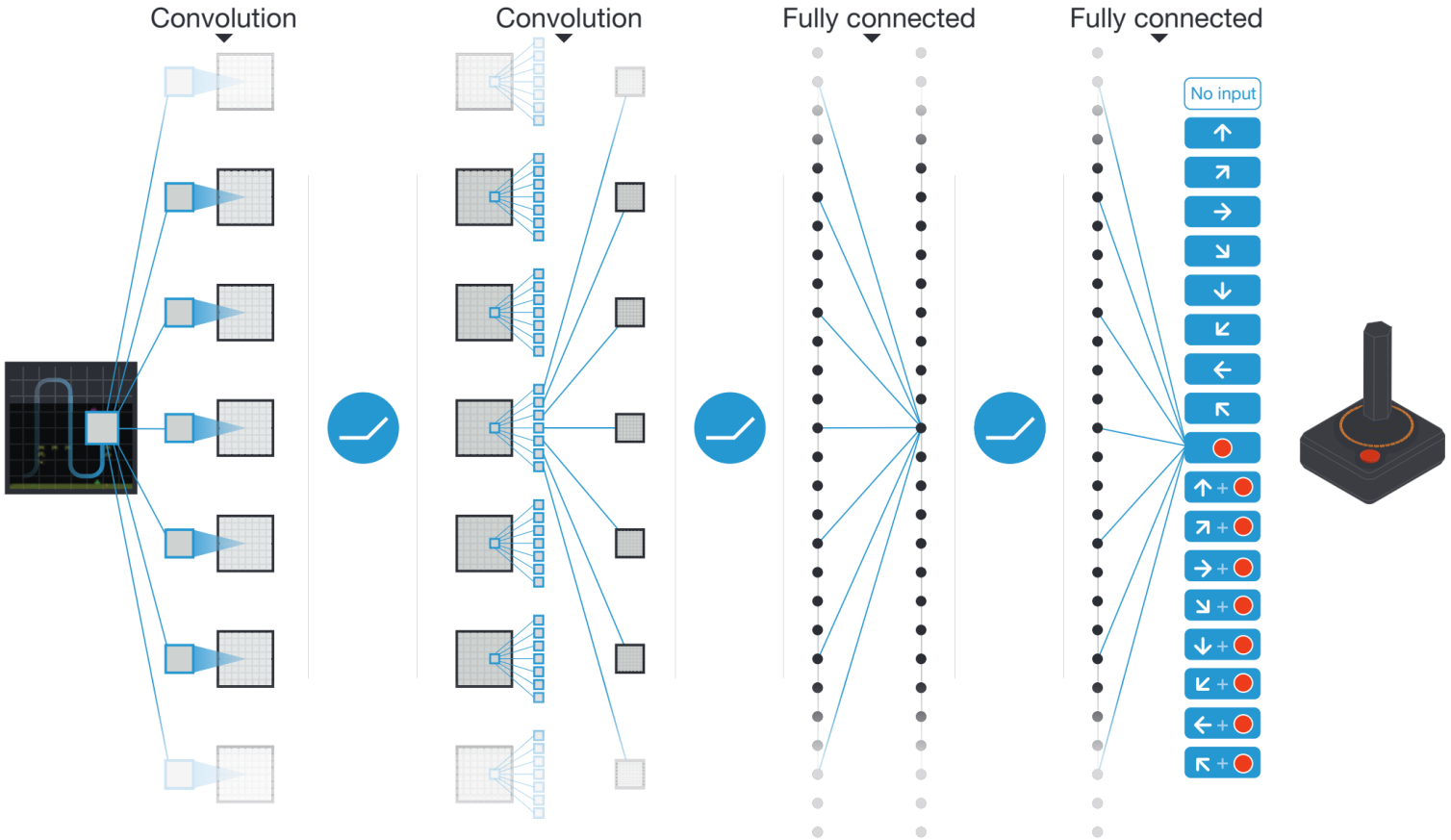
1. Draw a random number $r \in [0, 1]$.
2. Choose action a as:

$$a = \begin{cases} \text{random} & \text{if } r < \epsilon \\ \arg\max_a Q(s, a) & \text{if } r \geq \epsilon \end{cases}$$

Higher ϵ chooses more exploration.

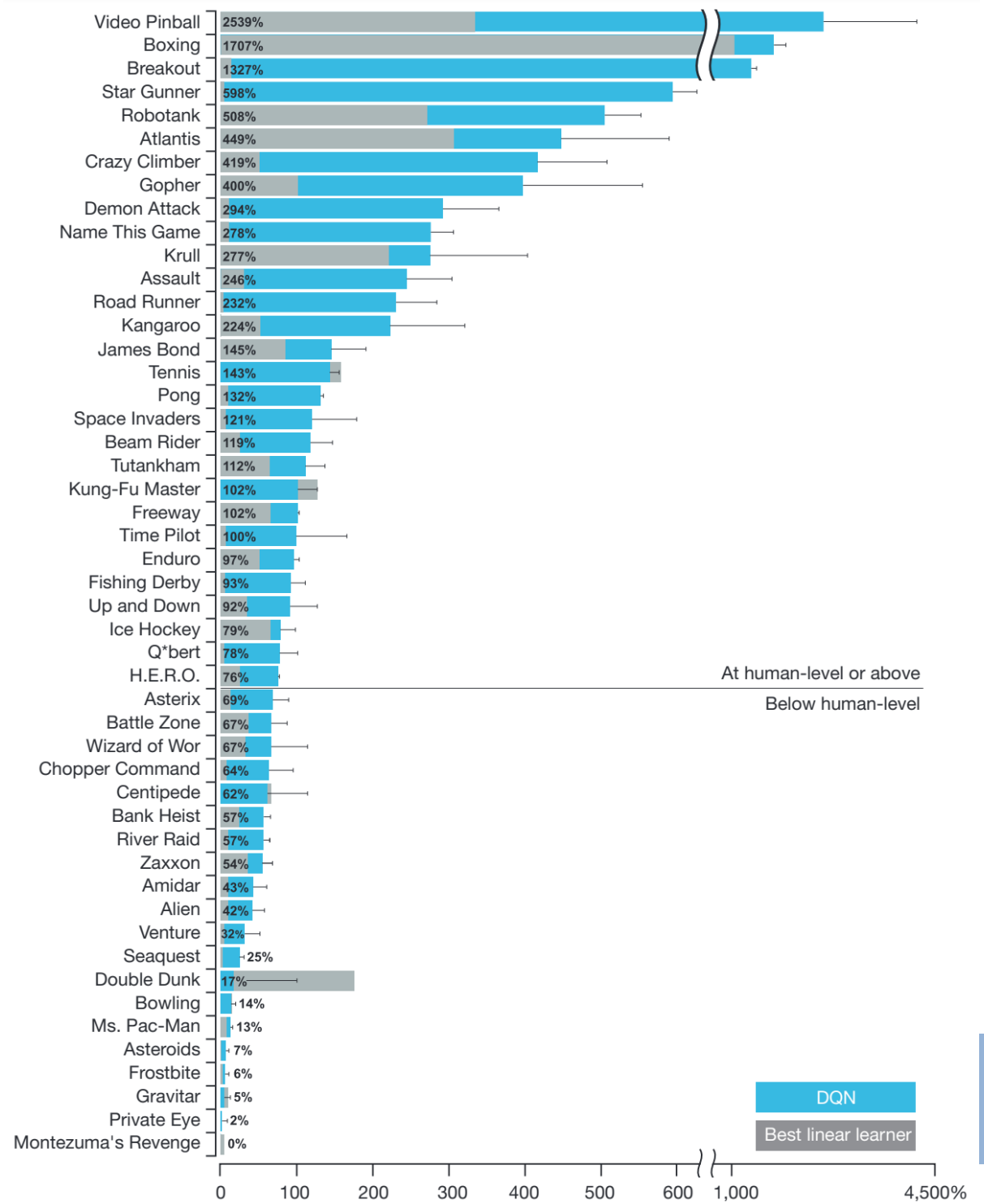
Deep Q Learning

Deepmind's work on ATARI games in 2015 launched work in Deep Reinforcement learning.



[Mnih et al., Human Level Control Through Deep Reinforcement Learning, Nature, 2015]

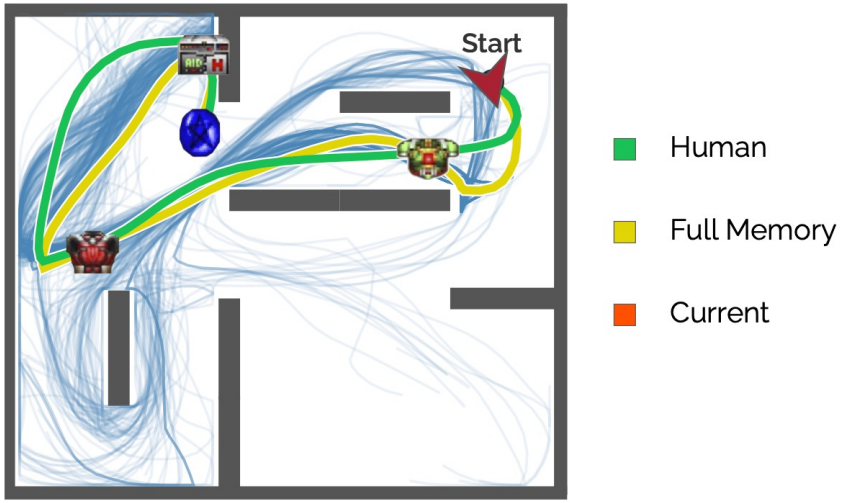
Google DeepMind's
Deep Q-learning

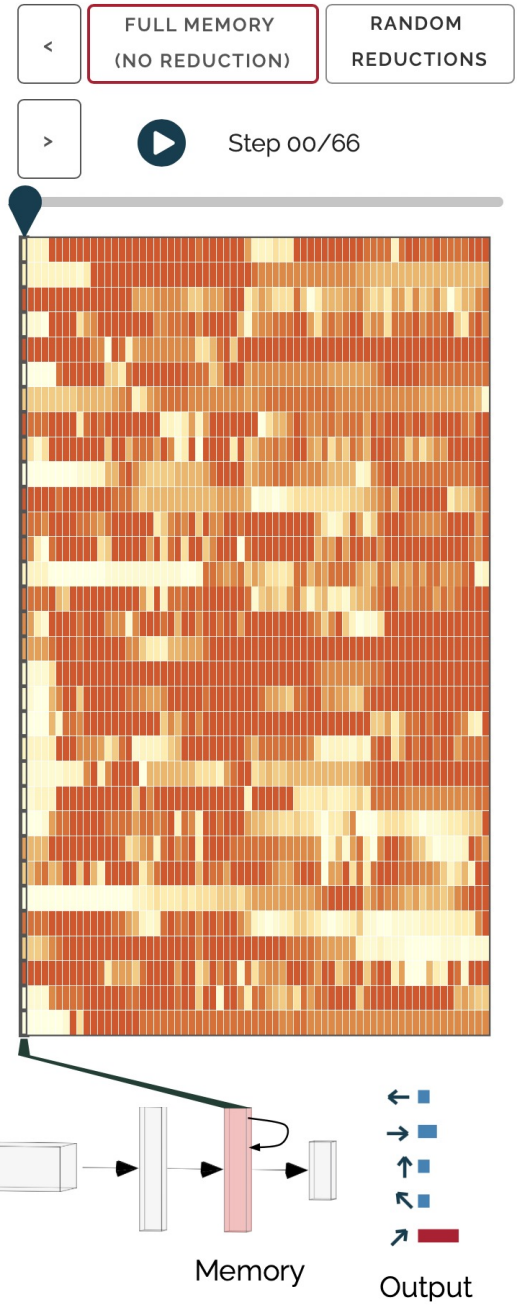


[Mnih et al., Human Level Control Through Deep Reinforcement Learning, Nature, 2015]

What if we Reduce the Memory of an Artificial Doom Player?

We built a Doom player AI using Deep Reinforcement learning. While playing, it builds and updates an inner representation (memory) of what it sees from the game. This memory represents what the AI knows about the game, and is the root of each decision. Reducing the size of the memory, could help the player learning to complete its task and thus lower its training time and energy consumption footprint. In this scenario, the player has to gather items in a specific order: Green Armor → Red Armor → Health Pack → Soul-sphere, with the shortest path possible.

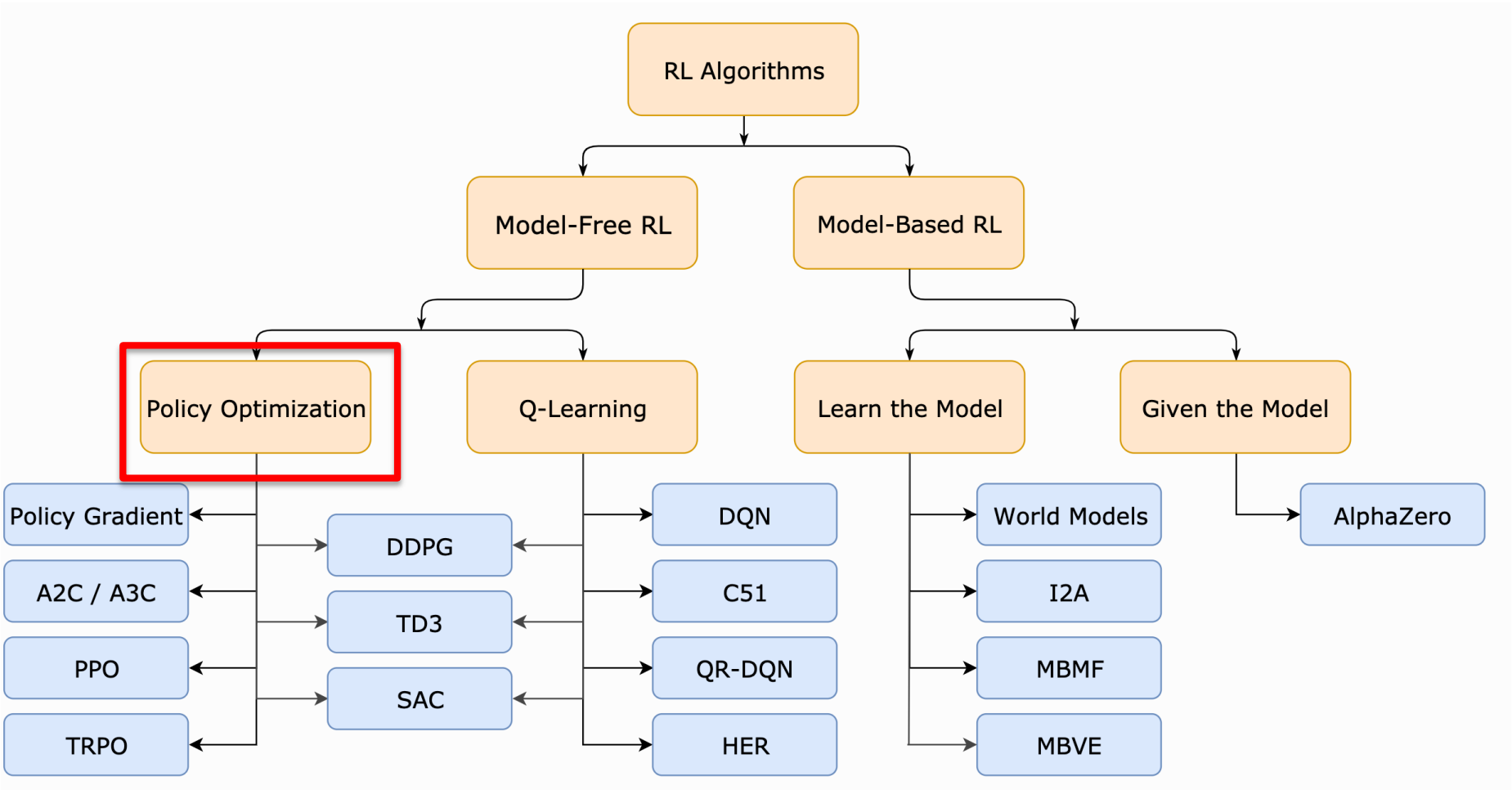




https://theo-jaunet.github.io/MemoryReduction/

Appendix:
Policy gradient

Types of reinforcement learning algorithms



https://spinningup.openai.com/en/latest/spinningup/rl_intro2.html

Policy Gradient

Policy gradient directly optimizes the expected cumulative reward over a parametrized policy function π_θ:

$$\begin{aligned} J(\pi_\theta) &= \mathop{\mathbb{E}}_{\tau \sim \pi_\theta}[R(\tau)] \\ &= \mathop{\mathbb{E}}_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T-1} r_{t+1} | \tau\right] \end{aligned}$$

We would like to learn the policy network π_θ with gradient descent:

$$\theta_{k+1} = \theta_k + \nu \nabla_\theta J(\pi_\theta)$$

How do we calculate the gradient $\nabla_\theta J(\pi_\theta)$?

Policy Gradient

$$
\begin{aligned}
\nabla_\theta J(\pi_\theta) &= \nabla_\theta \mathop{\mathbb{E}}_{\tau\sim\pi_\theta}[R(\tau)] \\
&= \nabla_\theta \int_\tau P(\tau|\theta)R(\tau) \\
&= \int_\tau \nabla_\theta P(\tau|\theta)R(\tau) \\
&= \int_\tau P(\tau|\theta)\nabla_\theta \log P(\tau|\theta)R(\tau) \\
&= \mathop{\mathbb{E}}_{\tau\sim\pi_\theta}[\nabla_\theta \log P(\tau|\theta)R(\tau)]
\end{aligned}
$$

$$
\nabla_\theta J(\pi_\theta) = \mathop{\mathbb{E}}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t)\, R(\tau)\right]
$$

https://spinningup.openai.com/en/latest/spinningup/rl_intro3.html